Demo Title

Lightwell Autonomous Remediation: Multi-Agent AI on Red Hat AI & Red Hat Advanced Developer Suite

Demo Abstract

Maintaining secure software supply chains requires rapidly validating and applying dependency patches when security platforms flag vulnerabilities. Enterprise teams struggle with the heavy manual overhead of analyzing Lightwell vulnerability alerts, updating code, and validating production readiness without introducing breaking changes.

This demo showcases an end-to-end autonomous DevSecOps solution driven by **Lightwell** and powered by **Red Hat AI (OpenShift AI)** alongside **Red Hat Advanced Developer Suite**:

- **Lightwell Trigger & Developer Portal**: Lightwell vulnerability alerts automatically initiate continuous remediation workflows directly inside **Red Hat Developer Hub**, cataloging security updates in real time.
- **Autonomous Multi-Agent Engine**: Hosted on **Red Hat AI**, dual-reader models (via MaaS) collaborate with an LLM-as-a-Judge to analyze Lightwell alerts and perform consensus-driven code patching.
- **Enterprise Guardrails**: Agent execution is safely contained using Agent Sandbox (OpenShell proxy filters and Landlock system controls) to ensure secure filesystem and network isolation.
- **Automated Validation**: Patched code undergoes automated unit, end-to-end, and smoke tests, with evaluation metrics logged in MLflow and EvalHub before submitting a production-ready Pull Request.

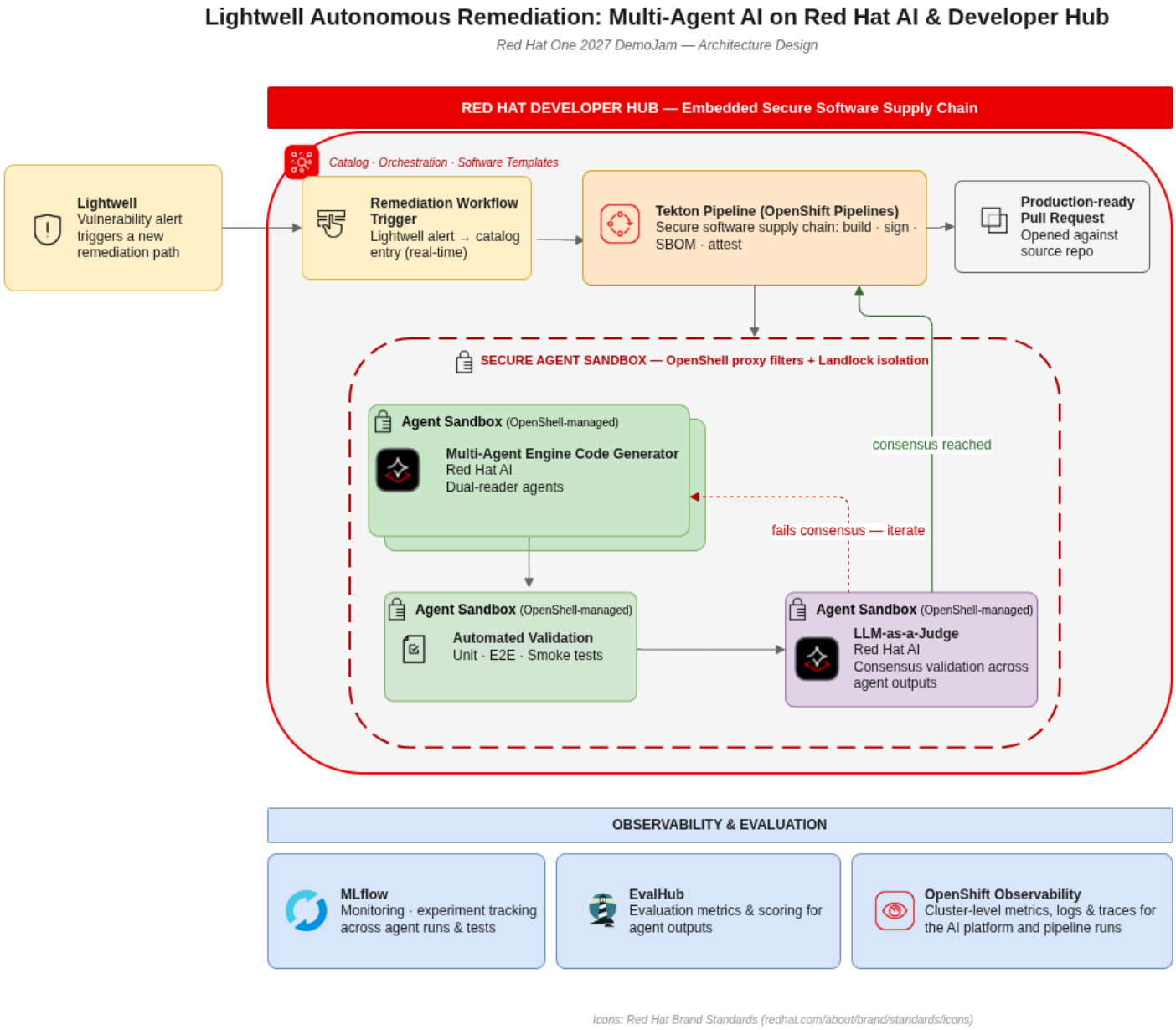
What is the business value proposition?

Enterprise customers rely on Lightwell for vulnerability detection but face massive backlog bottlenecks trying to remediate issues manually. This proposal transforms Lightwell from a passive detection tool into an active, self-healing DevSecOps workflow:

- **Accelerated Remediation (MTTR)**: Turns Lightwell vulnerability findings into validated, production-ready code fixes in minutes rather than days.
- **Zero-Trust AI Guardrails**: Solves the security risks of autonomous AI through sandboxed environments (Agent Sandbox/OpenShell) and consensus validation (LLM-as-a-Judge), ensuring autonomous agents cannot compromise systems.
- **Unified Developer Experience**: Eliminates developer patch fatigue while offering full visibility and compliance tracking across the software supply chain within Red Hat Developer Hub.

Additional Comments/Notes

- **Multi-Platform Alignment**: Directly links **Red Hat AI** (OpenShift AI, MLflow, Model Serving) with **Red Hat OpenShift & Red Hat Advanced Developer Suite** (Developer Hub, Software Supply Chain).
- **Replicability**: Designed for automated deployment via the Red Hat Demo Platform (RHDP) for global reuse across Tech Sales and Services.



Ideas:

- Pensar en el ejemplo para la demo, que biblioteca/s utizar y en que aplicación
    - El cambio de biblioteca supone cambio en codigo
    - El modelo es capaz de indentificar un test que falla antes del cambio de codigo
    - Genera el test y falla
    - Cambia el codigo
    - Lanza el test y funciona
- Integracion con DeveloperHub
- Reutilizar un pipelines devsecops
- Crear entornos de remediacion con Agent Sandbox y OpenShell inicialmente utilizando MaaS de Red Hat en integrado con MLFlow
- Realizar evaluacion para pero no tengo muy claro de que
- Automatizar que se lance el pipeline cuando una biblioteca nueva que afecta a una aplicacion

Repos que pueden ser interesantes:
https://github.com/RedCupofJoe/acs-ai-overwatch